

Copula Modelling for Mixed Continuous and Binary Variables in Survival Analysis: Applications to Colorectal Cancer

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Motivation

- ▶ Traditional survival analysis usually focuses on the time until a single event. (Kaplan-Meier estimator, Cox proportional hazard model and accelerated failure time models)
- ▶ Clinical trials often track multiple competing events for a single patient and we want to analyse and quantify the relationship between these events, and their effect on the survival probability.
- ▶ A copula-based joint survival function provide a way of quantifying the underlying dependence between competing events which current methods do not do.
- ▶ This presentation will cover the theory of survival analysis and copulas, as well as their application to a colorectal cancer data set.

Copula Theory

Definition

A d -dimensional copula $C : [0, 1]^d \rightarrow [0, 1]$ is a CDF with uniform marginals which we write $C(\mathbf{u}) = C(u_1, \dots, u_d)$ and satisfies:

1. $C(u_1, \dots, u_d)$ is non-decreasing in each component,
2. the i^{th} marginal distribution is obtained by
 $C(1, \dots, 1, u_i, 1, \dots, 1) = u_i$,
3. for $a_i \leq b_i$, $\mathbb{P}(U_1 \in [a_1, b_1], \dots, U_d \in [a_d, b_d]) \geq 0$. This is the rectangle inequality.

Sklar's Theorem

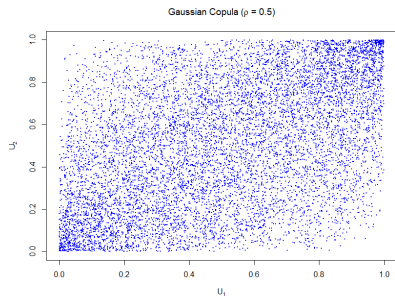
For a d -dimensional cumulative distribution function F , with marginals $F_1, \dots, F_d$, there exists a copula C such that

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)),$$

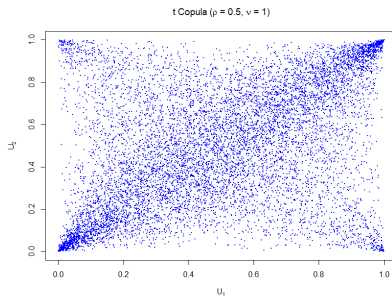
for all $x_i \in [-\infty, +\infty]$ and $i = 1, \dots, d$.

Elliptic Copulas

Gaussian and t copulas are types of elliptic copulas. They are symmetric with either no tail dependence or upper and lower tail dependence.



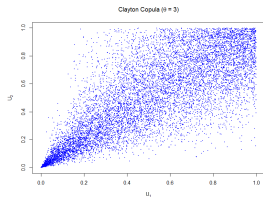
Gaussian copula ($\rho = 0.8$)



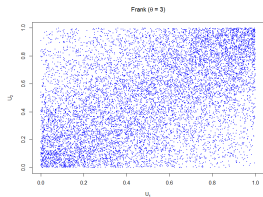
t-copula ($\rho = 0.5, \nu = 1$)

Archimedean Copulas

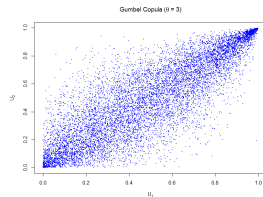
Clayton, Frank and Gumbel copulas are Archimedean copulas, defined by a single parameter θ and characterised by their tail dependence: Clayton (lower), Frank (none), Gumbel (upper).



Clayton ($\theta = 3$) -
lower tail



Frank ($\theta = 3$) - no
tail dependence



Gumbel ($\theta = 3$) -
upper tail

Survival Functions

The univariate survival function is given by

$$S(t) = \mathbb{P}(T > t) = 1 - F(t)$$

The joint survival function given by the survival copula $\tilde{C}$ is given by

$$S_{T_1, T_2}(t_1, t_2) = \tilde{C}(S_1(t_1), S_2(t_2))$$

We work with survival probabilities $S(t)$, not CDFs, so we need the survival copula in terms of the base copula. The relationship between the survival copula and base copula C is given by

$$\tilde{C}(u, v) = u + v - 1 + C(1 - u, 1 - v)$$

Copula-based Survival Estimator

The copula-based survival estimator is given by

$$S_{PFS}(t) = \hat{S}_{KM}^{(D)}(t) + \hat{S}_{KM}^{(Y)}(t) - 1 + C(1 - \hat{S}_{KM}^{(D)}(t), 1 - \hat{S}_{KM}^{(Y)}(t); \hat{\theta}),$$

where marginals, $\hat{S}_{KM}(t)$, is the Kaplan-Meier estimator.

$$\hat{S}_{KM}(t) = \prod_{t_m \leq t} \left(1 - \frac{d_m}{r_m}\right).$$

The inference steps are:

1. Fit a KM estimator to the new lesion and the tumour progression events which define the marginals.
2. Estimate the copula parameter θ via maximum likelihood estimation, which contains the dependence information between the marginals.
3. Substitute estimates into the equation for $S_{PFS}(t)$ to obtain an survival estimate for time t .
4. The bootstrap method is applied to obtain 95% confidence intervals.

Simulation Design

RECIST defines a progression event as

- ▶ a tumour size greater than 20% of the minimum of all previous tumour sizes.
- ▶ the appearance of one or more new lesions.

Single Arm Trial:

- ▶ 5 time points, $n = 150$,
 $\mu = (0.00, 0.036, 0.072, 0.108, 0.144)^T$
- ▶ 7 time points, $n = 150$,
 $\mu = (-0.22, -0.29, -0.36, -0.36, -0.43, -0.51, -0.36)^T$

Comparative trial:

- ▶ 200 patients/arm, 5 time points
- ▶ 10 scenarios varying treatment effect β , tumour-lesion dependence γ and tumour-size distribution (Gaussian vs. heavy-tailed t)

Single-Arm Simulation: 5 Time Points, $n = 150$

- ▶ All methods unbiased; point estimates match true $S(t)$
- ▶ Aug-boot: narrowest CIs when assumptions hold but slight under-coverage
- ▶ Gumbel Copula: narrower CIs than KM, coverage at or above nominal 95%

KM: 1000 iterations. Aug-boot, copula: 1000 iterations of 200 bootstrap samples.

Time	True $S(t)$	Coverage Rate			Avg. CI Width		
		KM	Aug-boot	Copula	KM	Aug-boot	Copula
1	0.587	0.960	0.943	0.965	0.193	0.123	0.189
2	0.329	0.949	0.945	0.959	0.186	0.109	0.178
3	0.213	0.949	0.931	0.958	0.163	0.088	0.153
4	0.115	0.961	0.936	0.960	0.130	0.060	0.118
5	0.063	0.980	0.925	0.992	0.103	0.039	0.095

A similar trend was identified in the 7 time point study, suggesting this method can be extended to longer studies with different means and covariance structures.

Comparative Trial: Power Across 10 Scenarios

- ▶ **Scenarios 1–4 (no dependence):** Aug-delta most powerful; copula competitive with KM
- ▶ **Scenarios 5–7 (tumour–lesion dependence):** copula matches or exceeds Aug-delta
- ▶ **Scenarios 8–10 (heavy-tailed tumour dynamics):** Aug-delta's Type I error inflates to $\sim 17\%$; copula stays at nominal 5% due to Gaussian assumptions being violated.

Power: 200 patients/arm, 1000 iterations, 200 bootstrap samples.

Scenario	Description	Log-rank	Aug-delta	Copula
1	True null (Type I)	0.040	0.045	0.045
2	No γ , small trt effect	0.371	0.405	0.395
3	No γ , medium trt effect	0.718	0.795	0.693
4	No γ , large trt effect	0.912	0.964	0.828
5	Differing γ , no trt effect	0.622	0.694	0.674
6	Differing γ , small trt effect	0.974	0.990	0.993
7	Differing γ , medium trt effect	0.996	0.999	0.998
8	Heavy-tail null, dep. misalign.	0.055	0.174	0.046
9	Heavy-tail H_1 , dep. misalign.	0.924	0.864	0.875
10	Heavy-tail null, matched dep.	0.057	0.170	0.044

Description of the Colorectal Cancer Data Set

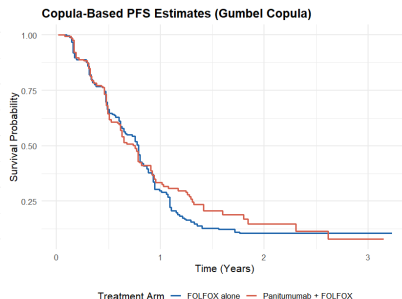
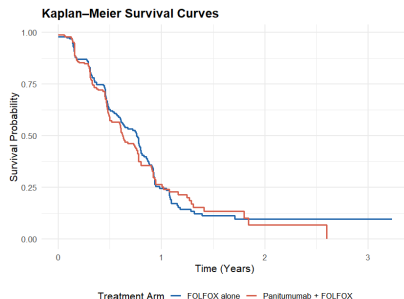
Description

- ▶ The trial compares the treatment of colorectal cancer with the drug FOLFOX4 and FOLFOX4 + panitumumab.
- ▶ The trial contained 1183 patients randomised into two treatment groups.
- ▶ Only 442 patients were considered in the analysis.

Description of colorectal cancer data set.

Treatment	FOLFOX alone	FOLFOX + panit.
Patients enrolled	593 (50.1%)	590 (49.9%)
Patients for the analysis	223 (50.5%)	219 (49.5%)
Patients with Wild-type KRAS	132 (59.2%)	91 (40.8%)
Patients with Mutant KRAS	128 (58.4%)	91 (41.6%)
Deaths	164 (73.5%)	164 (74.9%)
Composite progression events	76 (34.1%)	67 (30.6%)
New lesion events	61 (27.4%)	52 (23.7%)
SLD progression events	20 (9.0%)	18 (8.2%)

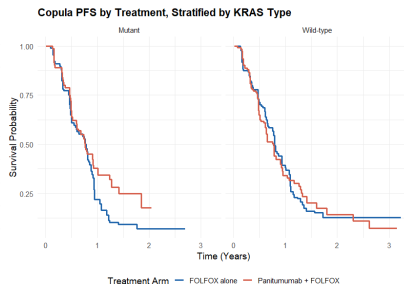
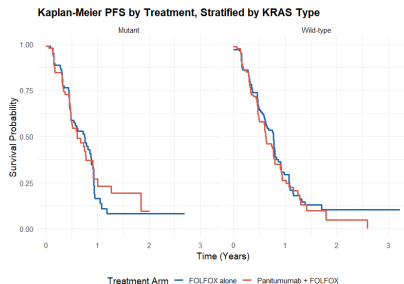
Data Analysis: Treatment Group



- ▶ Curves overlap throughout; no significant difference detected (log-rank $p=0.666$, copula L1 $p=0.343$).
- ▶ Both methods agree no separation between FOLFOX and FOLFOX+panitumumab in this subset.
- ▶ Kendall's $\tau = 0.26$ (FOLFOX) and 0.31 (FOLFOX+panit.)
- ▶ Moderate lower-tail dependence on the survival scale, meaning early tumour progression and early new lesions tend to co-occur

Data Analysis: KRAS Type

- $n=91/\text{arm}$, RECIST-only endpoint



The p-values for each KRAS type and estimation method.

Test	Wild-type	Mutant
Log-rank	0.455	0.879
Copula L_1	0.761	0.008

- Mutant has sustained 10pp separation between years 1–2.
- Log-rank under-weights late events; L_1 captures non-proportional hazards.
- Methodological finding, not a claim about panitumumab.

Limitations and Future Work

Limitations:

- ▶ Bivariate copula only; real trials involve more endpoints (e.g. death)
- ▶ Bootstrap + MLE is computationally expensive at scale
- ▶ Two-step inference process increases variance in the copula estimator and results in wider confidence intervals than Aug-bin
- ▶ PRIME analysis uses subset of 442/1183 patients with available longitudinal data

Future work:

- ▶ Multivariate copula extension
- ▶ Test whether lower tail dependence generalises across other RECIST trials
- ▶ Prognostic value of τ and tail dependence for subgroup stratification

Conclusions

- ▶ Copula-based joint survival estimator for mixed continuous/binary RECIST endpoints
- ▶ Competitive efficiency vs. KM and aug-bin, with robustness to non-Gaussian tumour dynamics where aug-bin's Type I error breaks
- ▶ Extra clinical inference via Kendall's τ and tail dependence; in PRIME, detects late KRAS-mutant survival divergence missed by log-rank